

CAM –EXAM 10/09/2021

SOLUTIONS TO NUMERICAL PROBLEMS

METAL FORMING

A cylindrical sample made of aluminum alloy ($K = 240 \text{ MPa}$, $n = 0,16$) with an initial height $h_0 = 100 \text{ mm}$ and an initial diameter $d_0 = 80 \text{ mm}$, is compressed between two flat dies.

- 1) Determine the height and the diameter of the sample if the true strain is equal to $\varepsilon = -0,1$.
- 2) Consider a deformation process that requires two steps: a first one to a height $h_1 = 90 \text{ mm}$, followed by a second step to a height $h_2 = 80 \text{ mm}$. Neglecting friction, determine the maximum force and ideal work required for each step of the deformation.

Solution

- 1) Determine the height and the diameter of the sample if the true strain is equal to $\varepsilon = -0,1$.

Considering the volume as constant:

$$V_0 = \frac{\pi d_0^2}{4} h_0 = V_f = \frac{\pi d_f^2}{4} h_f$$

Being the final height equal to:

$$\varepsilon = \ln \left(\frac{h_f}{h_0} \right)$$

$$h_f = h_0 e^\varepsilon = 100 \cdot e^{-0,1} = 90,48 \text{ mm}$$

The final diameter of the sample is equal to:

$$d_f = \sqrt{\frac{h_0}{h_f} d_0^2} = \sqrt{\frac{100}{90,48} 80^2} = 84,1 \text{ mm}$$

- 2) Consider a deformation process that requires two steps: a first one to a height $h_1 = 90 \text{ mm}$, followed by a second step to a height $h_2 = 80 \text{ mm}$. Neglecting friction, determine the force required for each step, and the ideal work spent for each step of the deformation.

Referring to the figure, the required forces are:

$$F_1 = \sigma_1 A_1$$

$$F_2 = \sigma_2 A_2$$

Where, the flow stresses are:

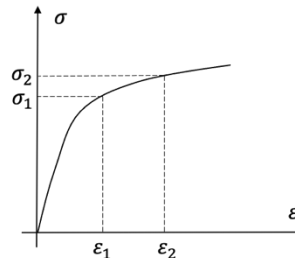
$$\sigma_1 = K \varepsilon_1^n$$

$$\sigma_2 = K \varepsilon_2^n$$

Being:

$$\varepsilon_1 = \ln \left(\frac{h_1}{h_0} \right) = \ln \left(\frac{90}{100} \right) = -0,105$$

$$\varepsilon_2 = \ln \left(\frac{h_2}{h_0} \right) = \ln \left(\frac{80}{100} \right) = -0,223$$



$$\sigma_1 = k\varepsilon_1^n = -240 \cdot (0,105)^{0,16} = -167,34 \text{ MPa}$$

$$\sigma_2 = k\varepsilon_2^n = -240 \cdot (0,223)^{0,16} = -188,77 \text{ MPa}$$

The volume is equal to:

$$V_0 = \frac{\pi 80^2}{4} 100 = 502.655 \text{ mm}^3$$

and the cross section of the specimen in the two steps equal to:

$$A_1 = \frac{V_0}{h_1} = \frac{502.655}{90} = 5.585 \text{ mm}^2$$

$$A_2 = \frac{V_0}{h_2} = \frac{502.655}{80} = 6.283 \text{ mm}^2$$

The required forces are:

$$F_1 = \sigma_1 A_1 = -167,34 \cdot 5.585 = -934.594 \text{ N} = -934,6 \text{ kN}$$

$$F_2 = \sigma_2 A_2 = -188,77 \cdot 6.283 = -1.186.042 \text{ N} = -1.186 \text{ kN}$$

The strain energy density in each step is equal to:

$$u_{0 \rightarrow 1} = \left| \frac{k\varepsilon_1^{n+1}}{n+1} \right| = \left| \frac{240 \cdot (-0,105)^{0,16+1}}{0,16+1} \right| = 15,15 \text{ MPa} = 0,0152 \frac{\text{J}}{\text{mm}^3}$$

$$u_{1 \rightarrow 2} = \left| \frac{k\varepsilon_2^{n+1}}{n+1} \right| - \left| \frac{k\varepsilon_1^{n+1}}{n+1} \right| = \left| \frac{240 \cdot (-0,223)^{0,16+1}}{0,16+1} \right| - 15,15 = 36,29 - 15,15 = 21,14 \text{ MPa} = 0,0211 \frac{\text{J}}{\text{mm}^3}$$

Therefor the work required is equal to:

$$W_{0 \rightarrow 1} = uV_0 = 0,0152 \cdot \frac{502.655}{1000} = 7,6 \text{ kJ}$$

$$W_{1 \rightarrow 2} = uV_0 = 0,0211 \cdot \frac{502.655}{1000} = 10,6 \text{ kJ}$$

TURNING

A steel bar with a diameter of 100 mm and a length of 300 mm is held by means of a self-centering three-jaw chuck and tailstock (dead centre). The workpiece undergoes external cylindrical turning, which brings the diameter to the final value of 94 mm over a length of 180 mm. The machining is performed in two steps: a roughing step and a finishing step. Considering the roughing step, whose depth of cut is equal to 2,5 mm, determine:

- 1) The values of cutting force and cutting power, taking into account a value of $k_{c0,4}$ equal to 2100 MPa, a coefficient x of the Kronenberg relationship equal to 0,29 and a value of main entering angle equal to 45° . The recommended feed is 0,15 mm/rev while the cutting speed is 260 m/min.
- 2) The optimal tool life and the corresponding cutting speed in case you want to optimize the production cost. The tool changing time is equal to 160 seconds, the cutting-edge cost is equal to 1 €, and the operating cost of the machine tool is 180 €/h, while the constants of the Taylor relation are $C = 250$ and $n = 0,122$.

Solution

Considering the roughing step, whose depth of cut is equal to 2,5 mm, determine:

- 1) The values of cutting force and cutting power, taking into account a value of $k_{c0,4}$ equal to 2100 MPa, a coefficient x of the Kronenberg relationship equal to 0,29 and a value of main entering angle equal to 45° . The recommended feed is 0,15 mm/rev while the cutting speed is 260 m/min.

$$k_c = k_{c0,4} \left(\frac{0,4}{f \cdot \sin k_r} \right)^x = 2100 \left(\frac{0,4}{0,15 \cdot \sin 45^\circ} \right)^{0,29} = 3.086 \text{ MPa}$$

$$A_D = f \cdot a_p = 0,15 \cdot 2,5 = 0,375 \text{ mm}^2$$

$$F_c = k_c A_D = 3.086 \cdot 0,375 = 1.157,25 \text{ N}$$

$$P_c = F_c v_c = 1.157,25 \cdot \frac{260}{60} = 5.015 \text{ W}$$

- 2) The optimal tool life and the corresponding cutting speed in case you want to optimize the production cost. The tool change time is equal to 160 seconds, the cutting-edge cost is equal to 1 €, and the operating cost of the machine tool is 180 €/h, while the constants of the Taylor relation are $C = 250$ and $n = 0,122$.

$$T_{opt}(T_c) = \left(T_t + \frac{C_t}{C_0} \right) \left(\frac{1}{n} - 1 \right) = \left(\frac{160}{60} + \frac{1}{180/60} \right) \left(\frac{1}{0,122} - 1 \right) = 21,59 \text{ min.}$$

$$v_{c,opt}(T_c) = \frac{C}{T_{opt}^n} = \frac{250}{21,59^{0,122}} = 171,85 \frac{\text{m}}{\text{min.}}$$

MILLING

A planar surface of a steel workpiece ($k_{cs} = 2800$ MPa, Kronenberg coefficient $x = 0,2$) must be machined. The planar surface has a rectangular shape with a length of 1000 mm and a width of 50 mm. The peripheral milling operation involves a roughing step characterized by a machining allowance equal to 5 mm. The selected tool has a diameter of 63 mm, a working height of 110 mm, and 4 cutting edges. The process parameters used are so that the feed velocity is equal to 480 mm/min, and the number of revolutions per minutes is equal to 600.

Determine the chip thickness, the cutting force, the cutting torque, the cutting power, and the cutting time, verifying that the operation can be performed with a milling machine with a power of 60 kW and characterized by an efficiency of 0,9.

Solution

To answer, it is necessary to understand what the actual situation in this slab milling operation is.

First, we need to compute the engagement angle and then the number of simultaneously working teeth:

$$\varphi = \arccos \frac{\frac{D}{2} - a_e}{\frac{D}{2}} = \arccos \frac{\frac{63}{2} - 5}{\frac{63}{2}} = \arccos 0,84 = 32,86^\circ = 0,574 \text{ rad}$$

$$\varphi_0 = \frac{2\pi}{Z} = \frac{2\pi}{4} = 1,571 \text{ rad} = 90^\circ$$

$$z = \frac{\varphi}{\varphi_0} = \frac{0,574}{1,571} = 0,37$$

Therefore, we must consider the worst-case scenario when a single cutting edge is cutting the maximum chip thickness, that we have when the angular position of the cutting edge is equal to the engagement angle.

To do the computation, we need to determine the feed per tooth from the data.

$$n = 600 \text{ RPM}$$

$$v_f = 480 \frac{\text{mm}}{\text{min}}$$

$$v_f = Z \cdot f_z \cdot n \rightarrow f_z = \frac{v_f}{Z \cdot n} = \frac{480}{4 \cdot 600} = 0,2 \frac{\text{mm}}{\text{rev}}$$

Therefore:

$$h_{D,max} = h_D(\theta = \varphi) = f_z \sin \varphi = 0,2 \cdot \sin 32,86^\circ = 0,109 \text{ mm}$$

$$A_{D,max} = A_D(\theta = \varphi) = h_{D,max} \cdot a_p = 0,109 \cdot 50 = 5,45 \text{ mm}^2$$

$$k_{c,max} = k_{cs} h_{D,max}^{-x} = 2800 \cdot 0,109^{-0,2} = 4.362 \text{ MPa}$$

$$F_{c,max} = k_{c,max} A_{D,max} = 4362 \cdot 5,45 = 23.773 \text{ N}$$

$$T_c = F_{c,max} \frac{D/2}{1000} = 23.773 \frac{63/2}{1000} = 748,8 \text{ Nm}$$

$$\omega = \frac{2\pi n}{60} = \frac{2\pi \cdot 600}{60} = 62,83 \frac{\text{rad}}{\text{s}}$$

$$P_c = T_c \omega = 748,8 \cdot 62,83 = 47.047 \text{ W} = 47 \text{ kW}$$

$$P_{c,limit} = E \cdot P_e = 0,9 \cdot 60 = 54 \text{ kW}$$

Therefore, being $P_c = 47 \text{ kW} \leq P_{c,limit} = 54 \text{ kW}$, the slab milling operation is feasible.

The cutting time in the slab milling operation is equal to:

$$t_c = \frac{L + \frac{D}{2} \sin \varphi}{v_f} = \frac{1000 + \frac{63}{2} \sin 32,86^\circ}{480} = 2,12 \text{ min.}$$